Do Inert Patterns Explain Sudoku Difficulty?

CHAERIN LEE

Mathematics–Statistics, Columbia University

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ABSTRACT. While solving Sudoku, one keeps checking patterns that turn out to change nothing. Checking them still costs attention. This study asks whether puzzles holding more such patterns take people longer.

The question is made precise, and it is shown that a pattern with nothing left to remove can never regain a target. The resulting measure is tested against human solving times for 138 puzzles, under a criterion fixed in advance. It fails. One control variable also turned out to be measuring a defect in the solver rather than a property of the puzzles. The conclusion is not that checking is free. It is that patterns a computer can enumerate were a poor stand-in for the ones a person stops to check.

1 Introduction

Sudoku difficulty ratings measure the road to the answer. They record the hardest technique a puzzle needs and how long the chain of reasoning runs. Pelánek tested such measures against human solving times and added a third, the dependency structure among steps [1]. All three matter.

None of these records the structures a solver might inspect and set aside.

A solver keeps meeting configurations that satisfy a rule exactly and accomplish nothing. Establishing that they accomplish nothing still takes time.

Puzzles may differ in how many such configurations they hold. This paper asks whether that difference explains solving time the conventional measures leave unexplained.

2 A Worked Example

A *candidate* is a number that could still go in an empty cell. As the puzzle is solved, candidates are crossed off.

Suppose two cells in the same box have the same two candidates, say 2 and 3. Between them they will use both digits. So no other cell in that box can hold a 2 or a 3. This is called a *naked pair*. Spotting one lets you cross those digits off everywhere else in the box.

Each example shows the puzzle as it stood at the step named, with one box shaded. That box is then repeated on its own, larger, with the possibilities that matter written in. Those cells hold other possibilities too. Only the ones the argument turns on are shown.

2.1 A pattern that does work. Puzzle 200, twenty-eight clues, sixteen players, mean solving time 478 seconds. After nineteen cells have been filled, the grid looks like this. The shaded box is on the left edge, in the middle band.

		6	1		8	9	7	4
			9		6		3	
			7	3	4			5
			5	4	9		1	2
4	1	5	3	7	2	8	6	9
9			6	8	1			7
6			2	9	3			
	9		4	1	5			
	5	1	8	6	7	4		3

Here is that box on its own.

3	3	3
4	1	5
9	2 3	2 3

The outlined cells can only be 2 or 3.
Between them they use both digits.
So the three threes above can go.

The pattern has three candidates it is entitled to remove, and all three are still there. Its *yield* is 3.

2.2 The same pattern, one step later. The solver did not use the pair. It made a different move: a 3 became forced at r4c7, over on the right of the same row. Placing it removed the 3 from every other cell in row 4, including all three shown above.

		6	1		8	9	7	4
			9		6		3	
			7	3	4			5
			5	4	9	3	1	2
4	1	5	3	7	2	8	6	9
9			6	8	1			7
6			2	9	3			
	9		4	1	5			
	5	1	8	6	7	4		3

The same box, at the same size and in the same place as before.

4	1	5
9	2 3	2 3

The outlined cells are unchanged.
The three threes have already gone.
There is nothing left to cross off.

The pattern is identical. Same two cells, same two digits, same rule satisfied. What changed is everything around it.

Recognising it now accomplishes nothing. This is the central observation of the paper:

> Useful and useless are not properties a pattern has. They describe what a pattern can still do at a particular moment.

We will call a pattern *productive* when it still has something to remove and *inert* when it does not.

2.3 A pattern that was inert from the start. Sections 2.1 and 2.2 show a pattern that was productive at one state and inert at the next, without ever being used. The quantity this study measures is different. It concerns patterns that had nothing to offer the first time they appeared.

Puzzle 431, twenty-six clues, ten players, mean solving time 306 seconds. After twenty-seven cells have been filled, the shaded box is the one in the bottom left corner.

8	5	4	9	2	3	7	1	6
6	2	3	5	7	1	9	8	4
7	9	1				5	2	3
	6	2		5	9	8		1
1	8	7			2		5	9
9		5	1	8		6		2
2	7			1				5
5	1			9		2		
			2		5	1		

2	7	
5	1	
3 4	3 4	

The outlined cells can only be 3 or 4.
Three other cells in the box are empty.
None of them could take a 3 or a 4.
When the pair first becomes visible,
there is already nothing to cross off.

Earlier deductions had already taken the threes and fours out of those three cells. The pattern is valid, it has somewhere it could act, and it removes nothing at the moment it appears. This is what the study counts.

3 From the Example to a Metric

The notation below is a short way of writing what Section 2 already showed. Nothing further is hidden in it.

3.1 Three definitions.

Definition 3.1 (*State*). A *state* S is a snapshot of the puzzle: the numbers already placed, together with the candidates still remaining in every empty cell. We write $C(S)$ for the set of all remaining candidates. Each grid printed above is a state.

Remark 3.2. No deduction rule is applied while building $C(S)$. Otherwise the patterns visible at a state would depend on how the snapshot was taken rather than on the puzzle.

Definition 3.3 (*Pattern instance and its targets***).** A *pattern instance* p consists of a rule together with the cells, the digits, and the unit that identify one occurrence of it. The structural requirement alone, without regard to whether anything can be removed, is called the *carrier condition*. Write $E(p)$ for the candidates the pattern would be entitled to remove if they were still present.

For the first example, p is the naked pair on digits 2 and 3 in the two highlighted cells of box 4. All four parts are needed. A naked pair on $\{3, 4\}$ in those same cells would be a different instance, with different targets.

There, $E(p)$ consists of every position in the rest of box 4 where a 2 or a 3 could ever be removed. Once the rule, cells, digits, and unit are fixed, $E(p)$ is fixed with them, and it does not change as the puzzle is solved.

Which of those positions still hold something is a separate question, and it is the one the state answers. At the state shown in Section 2.1, three of them do: the three threes in the top row of the box. That is what makes the next definition work.

The separation is deliberate. Some implementations require a naked pair to have something to remove before they report it. Under that reading the objects studied here do not exist. We use the weaker structural reading.

Definition 3.4 (*Yield***).** The *yield* of p at S is

$$\Delta(p, S) \;=\; |E(p) \cap C(S)|.$$

In words: of the candidates this pattern is allowed to remove, how many are still there. In Section 2.1 the answer is three, so $\Delta = 3$. One step later the same intersection is empty, so $\Delta = 0$. We call p productive when $\Delta(p, S) > 0$ and inert when $\Delta(p, S) = 0$.

3.2 Once inert, always inert. Solving a Sudoku only ever removes candidates. A digit crossed off a cell never comes back.

So a pattern with nothing left to remove can never acquire something to remove.

Lemma 3.5 (*Monotonicity***).** *Let p satisfy its carrier condition at states S_t and $S_{t'}$ with $t' > t$ along a solve. Then $\Delta(p, S_{t'}) \leq \Delta(p, S_t)$. In particular, $\Delta(p, S_t) = 0$ implies $\Delta(p, S_{t'}) = 0$ for every later t' at which p persists.*

Proof. Candidates are only removed, so $C(S_{t'}) \subseteq C(S_t)$. Since $E(p)$ is fixed by the rule, cells, digits, and unit defining p, it is the same set at both states, and $E(p) \cap C(S_{t'}) \subseteq E(p) \cap C(S_t)$. Taking cardinalities gives the inequality. If the right side is zero the left side is empty. ∎

Sections 2.1 and 2.2 show a pattern going from productive to inert. What matters there is not who removed the candidates, but that the same pattern's remaining yield fell from 3 to 0 as the state moved on. Lemma 3.5 says the reverse cannot happen.

The lemma is what allows each pattern to be recorded once, at the moment it first appears. A pattern inert on arrival stays inert. Counting it again at every later step would measure how long the puzzle ran, not how much dead structure it held.

3.3 The number. Along the observed path for Puzzle 431, the solver detected 115 distinct counted patterns. Of those, 66 were already inert the first time they became visible. So

$$\mathrm{IER} = \frac{66}{115} = 0.574,$$

About 57 percent of the patterns in that puzzle had no work left when they appeared. In general,

$$\mathrm{IER} = \frac{\text{patterns inert at first detection}}{\text{all distinct patterns detected}}.$$

Puzzle 200 gives $69/95 = 0.726$.

Neither number says on its own which puzzle is harder.

Singles are left out of the count. A visible single is almost always useful, so including them would turn the ratio into a measure of how many easy moves a puzzle offers.

Three other ways of counting were also tried. They are defined in Appendix A. All three agree with the main result.

4 Experimental Design

4.1 The guess. If checking a useless pattern costs attention, a puzzle full of them should take people longer. Two puzzles could need the same techniques at the same depth and still differ in how much dead structure lies along the way.

Puzzles with a high IER might be harder for ordinary reasons: denser candidate grids, harder techniques, longer solutions. Finding that they take longer would not show that IER was the cause. So the question was put differently:

> Given everything conventional we already know about a puzzle, does knowing its IER improve our prediction of human solving time?

Suppose one were predicting exam scores already knowing each student's study hours, attendance, and past grades. The question is not whether sleep correlates with scores. It is whether adding sleep improves the prediction once the other three are in hand.

4.2 Corpus, and its relation to development. The solver and the metrics were developed against 344 puzzles carrying human difficulty scores. Those were used to check the implementation, including the computational verification of Lemma 3.5. They were not used to choose the success criterion.

The test corpus is 138 puzzles from a public dataset of 1533 carrying average human solving times [2]. A puzzle was kept only if naked and hidden singles alone do not solve it. Every puzzle in the corpus therefore needs a counted rule.

The two sets overlap. 24 of the 138, or 17 percent, also appear in the development set. This is not a clean holdout. "Pre-specified" refers to the analysis plan, not to the sample.

The dependent variable is the natural logarithm of mean solving time in seconds. All correlations reported below are against that transformed variable.

4.3 Statistical procedure. The model is a linear regression. About each puzzle it is given five things: the clue count, the number of solving steps, the mean number of candidates per empty cell, the hardest technique required, and how many useful patterns are available per state. The question is what happens when IER is added as a sixth.

Adding any variable improves fit to the data a model was built on. That is arithmetic, not evidence. So the model was built on part of the corpus and tested on the puzzles it had not seen. Every figure below is that out-of-sample number. Exact settings are in Appendix B.

4.4 Criteria fixed before the test.

Criterion 4.1 (*Adequacy*). At least one control predictor must reach $|r| \geq 0.25$ against log mean solving time, the level reported for the weakest published baseline in [1]. Without this, a null result would say nothing about the hypothesis and only something about the corpus.

Criterion 4.2 (*Success*). The metric must improve cross-validated R^2 over the control model, with a coefficient keeping its sign across folds. Improvement in in-sample R^2 does not count.

5 Results

5.1 Adequacy, and one control that must be discarded. Three controls clear the threshold on their face: productive step count at $r = -0.332$, mean candidate count at +0.282, and highest required technique at +0.258.

The first is an artefact. It is discarded as evidence for adequacy, though it stays in the pre-specified control model of Table 1.

Its sign gave it away. More recorded steps went with *shorter* human times, which makes no sense. The fault is in the instrument. On harder puzzles the solver runs out of techniques and stops early, so it records fewer steps.

Step count correlates with solver completion at +0.876. Completed puzzles average 53.3 steps, stalled puzzles 26.3. Among completed puzzles alone the relationship with solving time disappears, at +0.067.

This is the error of measuring how hard a marathon was by the distance each runner covered before stopping. The runs that ended at ten kilometres look like the easy ones.

The other two survive, so the adequacy criterion is met without the contaminated predictor. The control model, unchanged from the plan, reaches cross-validated $R^2 = +0.152$.

5.2 The primary test. Every pre-specified metric improved fit to the data it was built on and made predictions worse on puzzles it had not seen. The three alternatives agree with the primary one.

The second half of the criterion behaves differently. The IER coefficient keeps its sign across all five folds under 17 of the 20 fold assignments. It is stable. The failure is in predictive value, not in instability.

Added metric	In-sample ΔR^2	Cross-validated ΔR^2
IER (primary)	+0.0070	−0.0067
Exposure-weighted ratio	+0.0002	−0.0236
Raw inert count	+0.0030	−0.0167
Expected random-order entry cost	+0.0117	−0.0040

Table 1. Incremental value over the control model, n = 138.

5.3 The same result seed by seed. A single mean can hide a split. Figure 1 gives the increment separately for each of the twenty fold assignments.

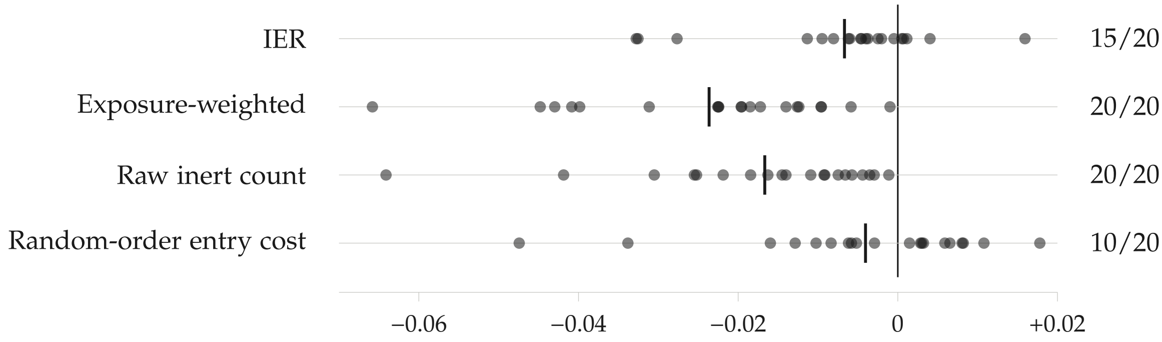

Figure 1. Change in cross-validated R^2 from adding each metric, one point per fold assignment. The vertical bar marks the mean. The count on the right is how many of the twenty fall below zero.

The picture is not uniform. Exposure-weighted ratio and raw inert count fall below zero under every assignment. IER falls below zero under 15 of 20. Random-order entry cost splits evenly at 10 of 20, with a median of −0.0007.

So the negative direction is robust to fold assignment for two of the four metrics and not for the other two. The twenty assignments are re-splits of the same 138 puzzles, not twenty fresh samples, so this shows stability under resampling rather than independent confirmation.

What holds across all four is the size. No single assignment of any metric produced an increment above +0.018, and most sit within a hundredth of zero.

5.4 Two specifications outside the pre-specified plan. Since step count proved to be an artefact, the control model was refit with a solver-completion indicator. It then reaches cross-validated R^2 = +0.1815, and adding IER gives +0.1824. The increment is +0.0009. Its sign is the reverse of the pre-specified model's, and its size is under one tenth of one percent of variance.

Among the 56 fully solved puzzles the control model reaches −0.1998, so it does not generalise at that sample size. Adding IER moves it to −0.2601. The correlation between IER and solving time there is +0.119.

Neither specification was fixed in advance. Neither is treated as a test.

Outcome. The first half of the success criterion is not met. The hypothesis failed its pre-specified test and is rejected as a predictive account for this corpus.

6 Scope and Limitations

6.1 What the study found, and what it did not. It found that adding IER did not meaningfully improve out-of-sample prediction of human solving time.

It did not find that human solvers experience no verification burden.

Those are different claims, because the study counted patterns a *program* enumerated. A person solving Puzzle 431 almost certainly never looked at that naked pair in box 7. The measurement may simply have failed to represent the thing it stood in for.

The idea was not shown to be wrong. The proxy built for it did not carry the weight.

6.2 Further bounds on the claim. The design has limited power. With 138 puzzles and a median of twelve players each, a small effect would not be visible.

The result is sensitive to specification. Under the pre-specified controls the increment is negative. Under a post-hoc model controlling for solver completion it is positive. A quantity that changes sign under a defensible respecification is not established in either direction. Even under the specification most favourable to IER, its mean increment was +0.0009.

Pelánek's dependency metric counts productive options, and the mirrored result was expected here. It did not appear. Mean productive availability reaches only $r = +0.132$ and does not clear the adequacy threshold.

6.3 Defects in the instrument. The solver fully solves 56 of the 138 puzzles. For the remaining 82 the metrics come from a truncated path, and puzzles that stall the solver earlier are observed over shorter paths. This is the most serious defect in the design. It disqualified one control outright, and it means every path-aggregated quantity here is observed unevenly across the corpus.

Highest required technique is defined by the solver and is therefore not independent of the instrumentation.

The counted rules are locked candidates, naked pairs, and hidden pairs. Triples, fish patterns, and chains are absent.

Median players per puzzle is twelve, against 131 and 1307 in [1]. Measurement error in the dependent variable attenuates every association reported here, including those of the controls.

One corpus, one platform, with the development overlap noted in Section 4.

7 Research Notes

The hypothesis began with repeated encounters with logically valid patterns that produced no elimination. Rather than treating that observation as evidence, it was treated as a hypothesis, and the result that would falsify it was specified in advance.

The hypothesis failed its pre-specified test, and it was rejected.

Appendix A. Three Alternative Counts

Writing $u(S)$ and $k(S)$ for the numbers of inert and productive patterns visible at S, the following are reported as robustness checks.

- *Exposure-weighted ratio*, $\sum_t u(S_t) / \sum_t (u(S_t) + k(S_t))$. This recounts a lingering inert pattern at every step it stays visible. It describes a solver who re-checks with no memory of having checked before.
- *Raw inert count*, the numerator alone, with no normalisation.
- *Expected random-order entry cost*, the mean over states of $u(S)/(k(S)+1)$. If patterns at S are examined in random order, each inert one comes before all $k(S)$ productive ones with probability $1/(k(S)+1)$. Summing over the inert patterns gives the expected number checked before the first useful one.

Appendix B. Statistical Detail

Models are ordinary least squares. Controls are clue count, productive step count, mean candidate count, highest required technique, and the mean number of productive patterns available per state. Highest required technique is encoded ordinally on the solver's three levels. Predictors are not standardised, which affects coefficient sizes but not fit statistics.

Five folds are used, with shuffling, regenerated under twenty random seeds numbered 0 through 19. The reported cross-validated R^2 is the mean across all folds and seeds. Within each fold, R^2 is $1 - \mathrm{SSE}/\mathrm{SST}$ on the held-out puzzles, with SST taken about the training-fold mean. Sign stability is reported as the number of the twenty fold assignments under which the coefficient keeps its sign in all five folds.

R^2 measures how much of the variation in solving time a model accounts for. The quantity of interest is not the total, but how much it moves when IER is added.

References

[1] R. Pelánek, *Difficulty rating of Sudoku puzzles: an overview and evaluation*, arXiv:1403.7373, 2014.

[2] S.-W. Wang, *A dataset of Sudoku puzzles with difficulty metrics experienced by human players*, IEEE Access **12** (2024), 104254–104262.

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